

Employee Attrition Prediction – Executive Summary

Prepared By:
ANJU KUMARI

Project Objective

The purpose of this project was to identify employees who are at a higher risk of leaving the organization before attrition occurs. Predicting employee attrition allows Human Resource (HR) teams to take proactive actions to improve employee retention, reduce hiring costs, and maintain workforce stability.

Key Findings

Analysis of employee data revealed several important patterns:

- Approximately **16% of employees** in the dataset left the organization, indicating that employee attrition is a significant business concern.
- The **Sales department** experienced the highest employee attrition rate.
- Among all job roles, **Sales Representatives** showed the highest likelihood of leaving the organization.
- Employees with **lower monthly income** generally showed higher attrition.
- Most employee exits occurred during the **first five years** of employment.
- The most influential factors affecting employee attrition were **Monthly Income, Age, Overtime, Total Working Years, and Number of Companies Worked**.

Business Recommendations

Based on the analysis, the following recommendations are suggested:

1. Focus on retention efforts on employees working in the Sales department, particularly Sales Representatives.
2. Monitor employees during their first five years of employment and provide career development opportunities.
3. Reduce excessive overtime and improve work-life balance to increase employee satisfaction.
4. Conduct regular employee engagement surveys to identify employees at risk before they decide to leave.

Conclusion

The developed machine learning model demonstrates that employee attrition can be predicted with good accuracy using employee and workplace-related information. The

insights generated from this project can help HR teams make proactive decisions, improve employee retention, and reduce organizational costs associated with employee turnover.